

Ship Security System

Onboard Ship Protection & Anti-Piracy Systems — Marine Engineering Notes

1. Introduction to Ship Protection

Ship protection refers to the combined set of physical, electronic and procedural measures taken to safeguard a vessel, her crew, cargo and passengers from threats such as piracy, armed robbery, stowaways, sabotage and unauthorised boarding. Modern merchant ships combine physical barriers, surveillance, detection, lighting and identification systems into a layered defence, in line with the ISPS Code (International Ship and Port Facility Security Code) and industry Best Management Practices (BMP) for piracy-prone areas.

Key Principles of Ship Protection:

- Deterrence — make the ship a difficult and unattractive target (lighting, wire, visible watchkeeping).
- Detection — identify a threat as early as possible (CCTV, radar, lookouts, AIS traffic awareness).
- Delay — slow down boarding attempts to buy time for response (razor wire, secured access, citadel).
- Defence & Response — alert the crew and authorities and follow the Ship Security Plan.

Governing Framework:

- ISPS Code — sets mandatory minimum security requirements for ships and port facilities.
- Ship Security Plan (SSP) — vessel-specific document detailing security measures and drills.
- Ship Security Officer (SSO) — crew member responsible for implementing the SSP.
- BMP (Best Management Practices) — industry guidance for transiting high-risk piracy areas.

2. Ship Razor Wire

Razor wire (also called concertina wire) is a physical anti-boarding barrier rigged around the vulnerable low-freeboard areas of a ship to prevent pirates or intruders from climbing aboard using grappling hooks, ladders or ropes.

- Rigged along the ship's rails, poop deck, stern, and other low-freeboard access points.
- Deployed in coils around the full perimeter of the weather deck in high-risk areas.
- Sharp-edged steel blades cause injury and snag ropes/hooks, discouraging climbing.
- Usually galvanised or stainless steel to resist rust and withstand weather at sea.
- Rigged in double or triple coil rows for stronger deterrence on high-risk voyages.
- Must be rigged and struck (removed) carefully, kept away from routine crew working areas.

3. Ship CCTV

Closed-Circuit Television (CCTV) systems give the bridge and security team continuous visual surveillance of the ship's deck, accommodation, engine room and access points, both for security and for general monitoring of operations.

Typical Coverage Areas:

- Weather deck perimeter, bow and stern.
- Accommodation entrances and stairways.

- Engine room and steering gear compartment.
- Cargo holds / cargo control areas.
- Bridge wings and pilot boarding points.

Key Features:

- Feeds are monitored and recorded at the bridge / security control station.
- Night-vision or infrared cameras used for low-light and night-time surveillance.
- Footage assists in verifying security incidents and post-incident investigation.
- Often integrated with the ship's alarm system to trigger recording on an alert.

4. Ship Alarm

Ship security alarm systems alert the crew, and in some cases shore authorities, of an unauthorised boarding, piracy attack, or other security emergency, prompting an immediate, pre-planned response.

Types of Alarms:

- Ship Security Alert System (SSAS) — a covert alert transmitted to the Company Security Officer and flag-state authority, without alerting an attacker on board.
- General/Emergency Alarm — audible ship-wide alarm used to muster the crew during an incident.
- Intruder/Perimeter Alarm — sensors on deck or access points that trigger when breached.

Response Procedure:

- Sound the alarm and muster crew per the Ship Security Plan.
- Proceed to the citadel (a secure, fortified space) if boarding is imminent.
- Activate SSAS to alert the Company Security Officer and flag administration.
- Maintain communication with naval/coastal authorities where possible.

5. Ship Upperdeck Lighting

Upperdeck (deck) lighting illuminates the ship's exterior at night, acting as a strong visual deterrent and aiding detection of approaching small craft or attempted boarding.

- Removes concealment for approaching skiffs or intruders during hours of darkness.
- Improves the effectiveness of CCTV and lookouts at night.
- High-intensity floodlights fitted along the ship's sides, stern and superstructure.
- Can be angled outward/downward to illuminate the waterline without blinding the bridge team.
- Used with a strict lighting policy set out in the vessel's BMP/security plan for high-risk waters.

6. Water Spray & Foam Monitoring

Water spray systems, and in some cases foam monitors, use the ship's fire main to create barriers of high-pressure water around the vessel's low-freeboard points, making it far harder for small skiffs to approach or for pirates to climb aboard.

- Creates a physical water curtain/barrier along the ship's sides, especially at the stern and quarters.
- Can be directed at approaching skiffs to swamp them or hamper boarding attempts.
- Doubles as part of the ship's existing fire-fighting system, requiring no separate dedicated hardware.
- Fixed or portable water monitors mounted around the weather deck.
- Can be remotely or manually activated from the bridge or deck stations.

- Considered a non-lethal, internationally accepted self-protection measure under BMP guidance.

7. Automatic Identification System (AIS)

AIS is an automatic tracking system that broadcasts and receives a ship's identity, position, course and speed, allowing other vessels, coastal authorities and vessel traffic services (VTS) to identify and track the ship. While primarily a safety and navigation tool, it also plays a security role.

Information Transmitted:

- Vessel name, call sign and IMO number.
- Position (GPS), course and speed.
- Ship type, dimensions and navigational status.
- Destination and estimated time of arrival.

Role in Security:

- Enables collision avoidance and traffic monitoring by coastal/naval authorities.
- Assists Search and Rescue (SAR) and maritime domain awareness.
- In high-risk piracy areas, ships may reduce AIS broadcast detail per BMP guidance to avoid revealing identifying information to potential attackers, while remaining compliant with SOLAS requirements.

8. Quick Reference Summary

System	Main Function
Ship Protection (Introduction)	Overall layered defence — deter, detect, delay, respond
Razor Wire	Physical anti-boarding barrier at low-freeboard points
CCTV	Visual surveillance and recording of deck and access points
Ship Alarm (SSAS)	Covert distress alert to Company Security Officer / flag state
Upperdeck Lighting	Night-time deterrence and visibility of the ship's exterior
Water Spray / Foam Monitoring	Repels small-craft boarding using the fire main
AIS	Vessel identification, tracking and traffic awareness

Note: Ship security measures must always be used in combination with the vessel's Ship Security Plan (SSP), crew training, and adherence to the ISPS Code and company Best Management Practices (BMP).